

# Sovin Herzlinger

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## EDUCATION

### California State Polytechnic University, Pomona

August 2023 – Present

Bachelor of Science in Electromechanical Systems Engineering Technology (EMSET)

Relevant coursework: Controls · Mechatronics · Introduction to Autonomous Vehicles

## WORK EXPERIENCE

### Early-Stage Hardware Startup

April 2025 – Present

Systems Engineer

- Designed and prototyped a device from the ground up — mechanical design, fabrication, firmware and software.
- Built out the engineering laboratory from scratch and contributed to core product research and development.
- Presented technical progress to stakeholders and led a small team of student engineers.

### Transcutaneous Spinal Stimulation Research

February 2026 – Present

Researcher | Casa Colina Hospital, Pomona, CA

- Assemble and calibrate a custom transcutaneous spinal cord stimulation research device, diagnosing and resolving hardware issues alongside a PhD researcher.
- Research targets stroke patients with disrupted motor cortex-to-spinal cord signaling; the objective is to detect residual neural impulses, amplify them, and re-emit stimulation directly to muscles to promote neural pathway regrowth.
- Improved device performance and data quality across experiments varying pulse duration, period, and amplitude. Preparing for clinical patient trials, Summer 2027.

## PROJECTS AND VENTURES

### Linear Inverted Pendulum — Self-Balancing Cart-Pole (Controls Lab)

September 2025 – Present

- Designed and built a belt-reduced linear inverted pendulum driven by an ODrive S1, with an AS5600 magnetic encoder at the pivot for high-resolution angle feedback.
- Architected a distributed CAN bus system across three nodes (two Teensy 4.x microcontrollers and the ODrive), separating sensor acquisition, real-time control, and motor drive into independent subsystems.
- Implemented an LQR stabilization controller running at 600 Hz across both nodes, catching and holding the pendulum upright from a hand-raised start.
- Wrote a Python LQR design tool (Riccati solve over the linearized plant) and validated it against hardware: every tuned gain landed within 30% of the model, the damping gain within 0.2%.
- Exposed live gain tuning over USB serial so all four LQR gains could be changed while balancing, enabling rapid iteration without re-flashing firmware.

### QuickCheck — Campus Club Check-In Platform

June 2025 – Present

- Founded and built a check-in, membership, and dues platform for campus organizations, now a native iOS app in external TestFlight beta; onboarded and managed a contracted engineer, handling contract negotiation, legal review, and scope.
- Rebuilt the product natively in React Native / Expo on a Supabase Postgres backend, enforcing authorization through row-level security policies in the database rather than in application code.
- Ran a security review against the live database and remediated every high-severity finding, verifying each backend fix as a real authenticated user.

### MotoNav — Motorcycle Navigation Display

2026

- Built a handlebar-mounted turn-by-turn display — an ESP32 driving a TFT in a 3D-printed mount, paired with companion iOS and Android apps that handle Mapbox routing and stream instructions over BLE — designed for glanceability at speed and validated by on-road testing.

### AEON — Self-Hosted AI Assistant & Home Server

2026

- Built a personal AI assistant on my own home server — a FastAPI service on a Raspberry Pi that takes real actions through a tool interface.
- Administer the Linux server it runs on: containerized services, private mesh networking with no public exposure, and automated integrity-checked backups replicated offsite.

## SKILLS

**Mechanical:** SolidWorks | Onshape | 3D Printing | Fabrication | Soldering | Prototyping

**Controls & Embedded:** LQR | PID | State-Space Modeling | CAN Bus | BLE | I<sup>2</sup>C | Teensy | ESP32 | ODrive | C++

**Software:** Python | TypeScript | React Native / Expo | Swift | FastAPI | PostgreSQL | Supabase | Linux | Git